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AI Action Plan
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After reading highly publicized submissions from our very-well funded competitors, I felt compelled to respond to the 'AI Action Plan' RFI. We bring a unique perspective that's quite different than the big tech companies and LLM startups in San Francisco fueled by billions of dollars of investment.

KYield is a true pioneer in AI systems, self-funded from the theorem stage nearly three decades ago. We have not enjoyed the publicity of our peers, so you may not have heard of us. I'll provide an introduction in the form of quotes from a letter I sent to President Trump during his first administration, several years prior to the launch of ChatGPT, which triggered the AI arms race.

"I am the founder of KYield and inventor of the KYield OS (KOS), which is a distributed AI operating system (OS). The foundational theorem *yield management of knowledge*.... posits that knowledge creation and transfer over distributed networks is a dynamic and complex undertaking consisting of.... properties with a malleable yield curve.... The KOS is now viable and timely for scaling due to recent exponential improvements in hardware and algorithmics."

"The now patented core of the distributed KOS includes a rules-based governance system, (applications) for business units or partners, and individual modules for each person (DANA). The standard KOS provides an automated approach to workflow augmentation (a.k.a. enhancement or amplification) functioning behind the scene throughout the networked organization, thus enabling a very dynamic and high-performance distributed AI OS. While each module is transparent in describing algorithmic type and function, specific details are withheld as trade secrets. Customers retain data control and ownership with the exception of limited metadata for underwriters and/or regulators per agreement."

I also sent letters during President Trump's first term to the Secretary of Defense, Joint Chiefs of Staff, Congress and the National Governor's Association warning about the risks from China in AI. We could see patterns over time that were quite disturbing, suggesting that China not only had more interest in our work than the U.S., but appeared to be investing significant resources in an apparent attempt to reverse engineer our work in AI systems.

These patterns led us to stop publishing any disclosures on our most advanced AI research surrounding the Synthetic Genius Machine (SGM). Indeed, I decided not to pursue the three patents recommended by the USPTO after my initial application for the SGM, but rather maintain the body of work as tightly guarded trade secrets (2019).

Once I fully realized that any system that can accelerate discovery can accelerate destruction, combined with the patterns in our web logs, we changed our policies. Unfortunately, most of my peers continue to immediately publish emerging AI research to the web, driven mostly by competitive pressure and incentives from Big Techs.

In contrast to many of our peers, rather than release advanced AI to the public prematurely, thereby unleashing significant risks, we made the choice to follow best practices in safety engineering required of other high-risk industries, including the handling of nuclear and biohazard materials. Our reward for doing the right thing in the current U.S. AI market has been severe financial punishment, whereas other companies in our industry have been rewarded with \$10s of billions of dollars for doing what leading AI scientists have [warned to be reckless](#), including causing existential risk.

In terms of private property and economics, Big Techs and their LLM partners are in the process of transferring the knowledge economy to their companies by training their models on data owned by others and monetizing it. The knowledge economy represents the majority of economic activity today in the U.S.

Policy makers should understand that strong monopolies and oligopolies don't need IP protection nearly as much as small potential competitors, which is presumably why Big Tech companies have worked to erode patents in the U.S. for the past two decades.

Therein provides a glimpse of the problems with U.S. AI policy. The greatest problem is the lack of any governance whatsoever beyond market power, resulting in a market where perverse incentives drive behavior, which determines outcomes, much of which is directly opposed to the interests of the U.S., its citizens, and the world.

Identifying, understanding, overcoming and preventing such perverse incentives in the future should be the priority for U.S. government policy in AI, as our following recommendations highlight.

Respectfully submitted,

Mark Montgomery
Founder & CEO
KYield, Inc.

Big Techs are driving U.S. AI policy. That needs to change.

We advocate for AI policy based on actual evidence, rather than conflicted actors and hype that often contains misinformation, as is currently the case in our industry.

Allow me to make some present-day comparisons to internal U.S. policy versus external. We are currently engaged in a trade war due to what the Trump Administration has described as unfair trade policies, resulting in a recent U.S. trade deficit of \$1.2 trillion. As is widely understood in business, some of the non-tariff policies are also very damaging, including government-wide subsidies for strategic industries.

For example, China's predatory pricing at far below cost are legendary, and cause damage to nations and industries worldwide, but their subsidies pale in comparison on a percentage basis to the subsidies occurring in the AI industry.

LLM startups that have raised billions of dollars, some of which are backed by the most valuable Big Techs, are [losing vast sums of money](#) in subsidizing customers, even though the majority aren't paying for the data their products are based on.

"They are, in effect, receiving subsidies of hundreds of billions of dollars. The United States will not be doing that any longer." President Trump's speech to a joint session of Congress on March 4th, 2025, referring to non-reciprocal tariffs.

If it were not for the [exceptionally large](#) Big Tech investments in LLM startups, these subsidies would not be possible. The intent of the subsidies employed by Big Tech, LLM startups, and the CCP are quite similar—to protect strategic interests, destroy competition, increase dependencies and then increase prices.

Our position is that U.S. policy should discourage predatory subsidies, and U.S. policies on subsidies should be the same whether domestic or international.

Obsession with scale

Another negative influence on U.S. markets and economy in AI is the obsession with scale in LLMs, which is also driven by the strategic interests of Big Techs. Big Tech has dominated investment in AI research since the [previous AI winter](#), which ended at the beginning of this century. The super majority of research that was commercialized by Big Techs has unsurprisingly been focused on the strategic interests of each company.

Conflicts in research is the primary reason we remained lean and self-funded at KYield for such an extended period—to allow us to follow the evidence to meet customer and societal needs rather than the needs of a conflicted sponsor. One commonality Big Tech companies share is an increasingly impenetrable scale ceiling. LLMs are viewed as a potential solution as they require more cloud computing customers pay extra for, hence the unprecedented strategic investments in LLMs and related infrastructure.

Unfortunately for society, LLMs have [inherent technical flaws](#), including unique catastrophic and cyber risks, the latter already being [realized at scale](#). Catastrophic risks such as accelerating bioterrorism will take more time to manifest as it requires a physical lab. However, based on nearly three decades of research at the confluence of human-caused catastrophes, AI systems, and prevention, I have little doubt such efforts have been underway since the first consumer LLM was unleashed to the public.

[Neurosymbolic AI](#) and other methods that focus on [distillation and compression](#) rather than scale are much more responsible than LLMs, but have been starved for funding due to the dominance of Big Tech in AI R&D. Many leading scientists have warned about the obsession with LLMs and AGI causing catastrophic risks, including existential.

Rather than supporting LLMs and Big Techs, the U.S. should be focused on increased security, protection of human rights, private property, economic and environmental sustainability—AI systems that are more consistent with the U.S. construct and values. Otherwise, we may lose leadership in technology as leaders with integrity may decide to locate in countries with more moral policies, fair competition, and functional markets.

Consolidation of wealth and power

Perhaps the most obvious trend in AI is the ultra-consolidation of wealth in a small group of companies headquartered in a few zip codes in the San Francisco and Seattle metro areas. One only needs to glance at the stock market for an indication of who investors perceive to be the primary beneficiaries of the AI/LLM boom, which is enabled by U.S. policies, or lack thereof.

At the end of 2024, the so-called mag seven companies represented 35.4% of the S&P 500's market capitalization, or nearly three times the percentage of a decade earlier, peaking at nearly \$20 trillion in market capitalization a few months ago. The enormous investment in the past two years is attributed to the expected growth in AI dominated by Big Techs. This concentration has resulted in warnings about systemic financial risk by central banks. We've been warning about related systemic risks for years.

Unlike previous generations of new technology—when current market leaders were small, lean startups, and I and others assisted, markets today are expecting incumbents to dominate AI markets. Big Techs are preventing others from having the opportunities provided to them, which I find unconscionable. Worse yet has been to observe the unhealthy relationships, increased government capture, and support for anti-market behavior by the U.S. government.

This megatrend of consolidation in market power and wealth is directly counter to the best interests of the U.S. and its citizens, and it's mostly due to the type of AI selected by Big Techs (LLMs), lack of timely response by the courts in copyright and antitrust lawsuits, and the lack of prudent policies by the U.S. government.

With a wealth gap bouncing along at record levels, wide-spread market failure, and a host of socioeconomic challenges, the U.S. should be aggressively pursuing rich diversification in AI technologies, companies and regions.

Conclusion

Prior to the LLM hype-storm and unprecedented predatory investment, the primary commonality most of us shared working in AI was recognition of the great potential to solve problems. Investment capital is necessary, and well-earned wealth creation welcomed, but the primary motivation was to accelerate discovery to cure diseases, prevent catastrophes, improve productivity to cure structural economic problems, and empower people to live healthier, happier lives.

Beginning with the unleashing of consumer LLMs to the public in November of 2022, which stunned many of us due to the known risks, AI was quickly repositioned to be just LLMs that are strategic vessels to protect and extend the market power and market capitalization of Big Techs, a pathway to instant wealth for AI scientists and engineers, and a target for pump and dump schemes by some unscrupulous investors.

Training on data owned by others for basic research was widely considered acceptable, but monetizing products trained on data owned by others was considered so unethical most wouldn't have considered it. I consider it to be the greatest theft in history. All of our rights are reflected in data today, including human rights, property rights and all other constitutional rights. If we don't have control of our data, rights are exploited.

Performing experiments with LLMs and other self-generating algorithms in labs was common, and didn't create problems for society as the risk was contained in a controlled environment, not unleashed to the world, including every evildoer on the planet. All that changed when LLMs were unleashed to the public.

Many of us find these trends very troubling, including quite a few of the most talented who made this boom possible with their intellectual contributions. If the U.S. government doesn't aggressively change course and establish responsible AI policies, I fear that it may result in the U.S. losing our leadership position in technology.

It is not the market power or market capitalization that creates or maintains leadership in science-related industries like AI, but rather the genius of a relatively few people, most of whom have high moral standards, and are careful who they empower.

I therefore recommend the following changes to U.S. AI policy with the goal of solving problems rather than creating them:

1. First, do no harm. Establish and require safety engineering principles for AI.
2. Create prudent AI regulations based on actual risk profiles. Self-generating models trained on vast data for the public is very high-risk and should be tightly regulated. Most other methods are low-risk and require little regulation.
3. Require a review process for publishing emerging research involving high-risk technologies. Invite and compensate expert reviewers from multiple disciplines with real-world experience, including national security, not just LLM scientists.
4. Make it a felony to knowingly introduce any type of high-risk technology to the public, and aggressively enforce it.
5. Leverage the entire U.S. government in collaboration with states to diversify AI rather than further consolidate one of the strongest oligopolies in history.
6. Reinforce human rights and private property rights, including an outright ban of training on data owned by others without an explicit license.
7. Instruct the SEC and other regulators to help rein in predatory subsidies and pump and dump behavior—it has a toxic influence on markets and building companies.
8. Lead a whole-nation approach to creative destruction, including reforming procurement systems that are often corrupted. Restore bias for the 'little guy'.

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